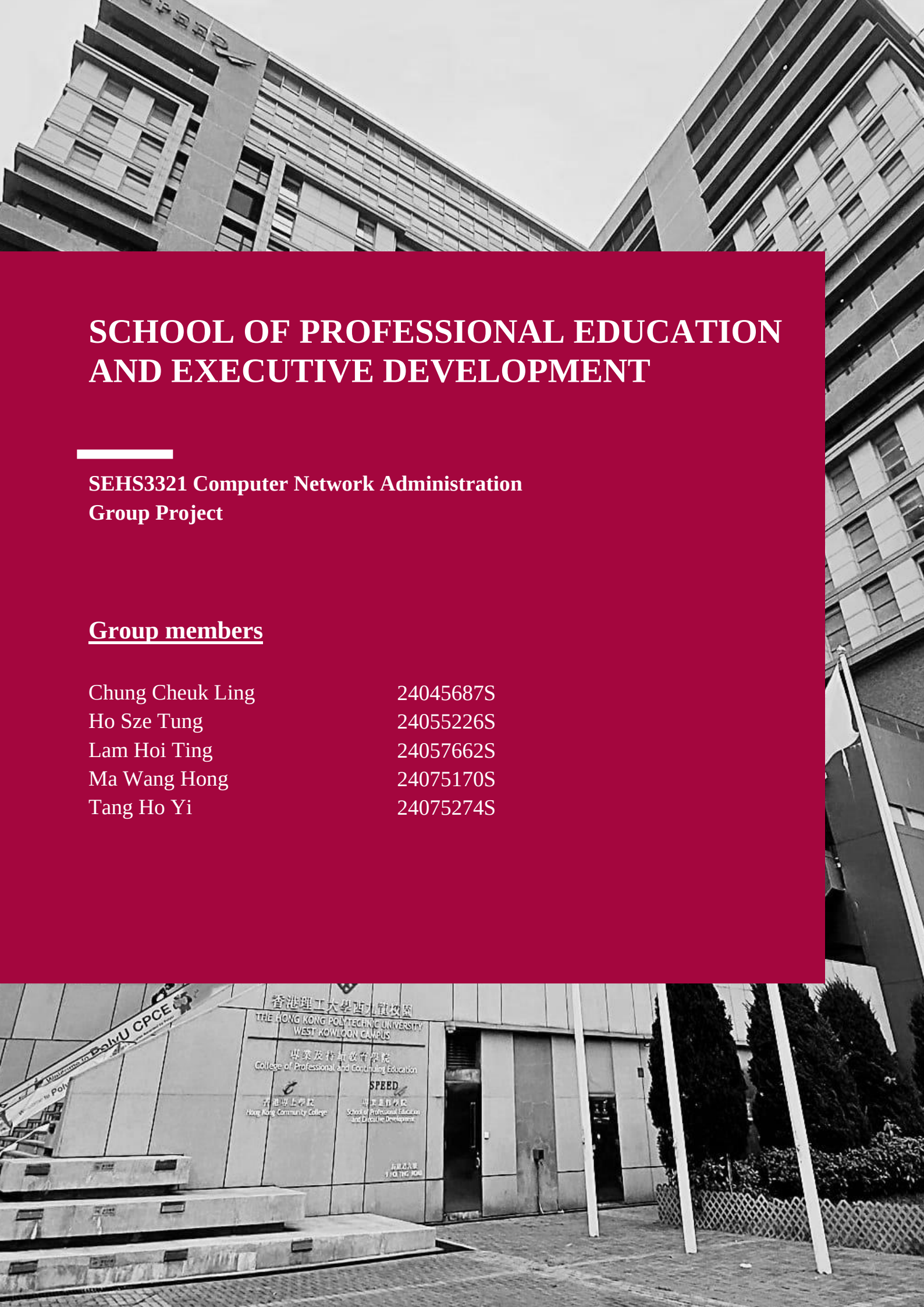




SCHOOL OF PROFESSIONAL EDUCATION AND EXECUTIVE DEVELOPMENT

**SEHS3321 Computer Network Administration
Group Project**

Group members

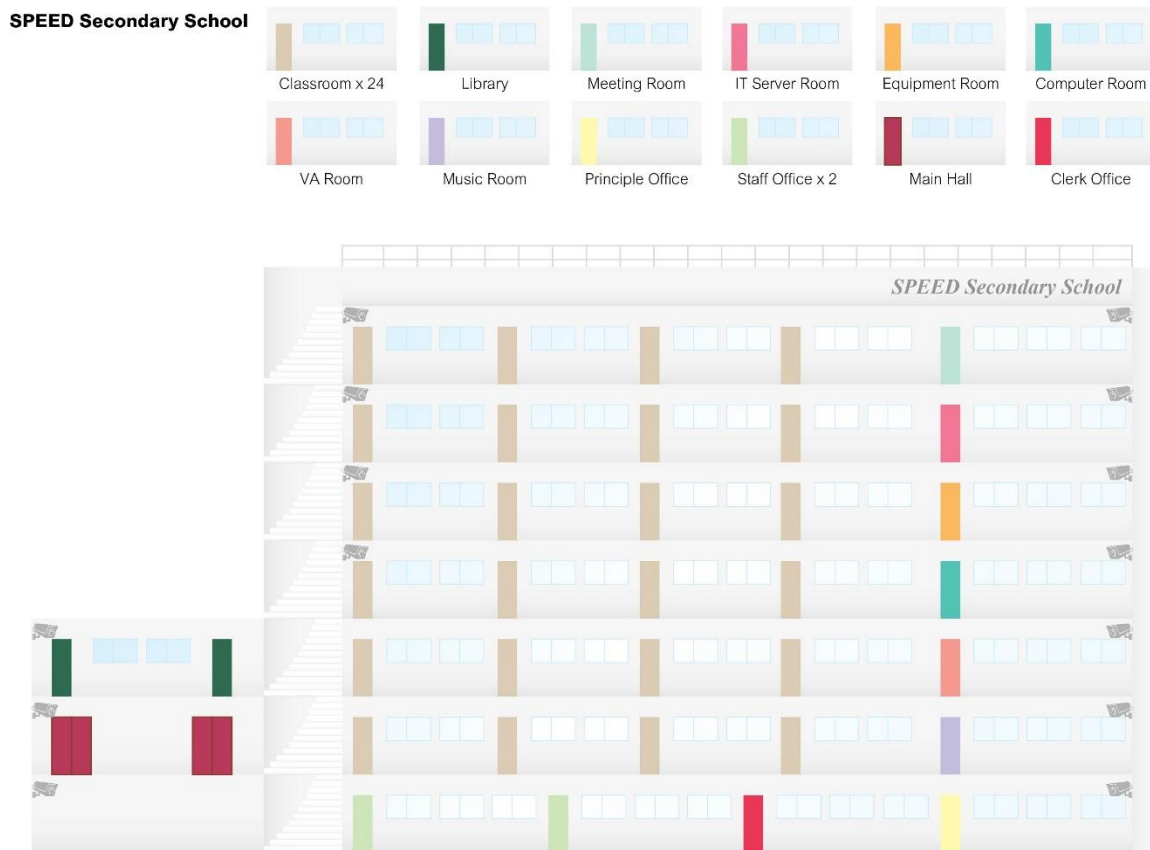
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Background

SPEED Secondary School is a newly established secondary international school. The campus requires a new network infrastructure for providing advanced and efficient digital services and enhancing network security. With 60 teaching staff and 10 administrative clerks, school provides different educations and activities from Secondary 1 to Secondary 6 with 4 classes in each grade. There are 7 floors with a rooftop in the school building. Principle, staff and clerk offices are on the ground floor. 4 classrooms on each floor. Main Hall and Library are on the 1st and 2nd floor. The equipment room, IT server room and Meeting Room are on the 4th to 7th floor. Each floor has 2 CCTV for students and staff safety. Please see the following figure for details.

School Floor Plan



Number of IT facilities

Facilities	CCTV	Telephone	Printer	Computer	Projectors	Servers	Total
Number	14	74	11	140	30	7	276

The network will be divided into 4 areas, classroom, staff room and office, library and IT room and wireless devices. There are around 290 devices that need connections which will arrange 4 subnets for school. Besides, setting up around 7 servers, web server, email server and DNS server, etc. and 2 WIFI devices for students and staff to use.

Network Design

Network Components

Each subnet contains a Cisco Catalyst 2950 Series 24-Port Fast Ethernet Switch to connect all the nodes. This switch is stackable which provides wire-speed Fast Ethernet connectivity. It is suitable for different network environments. Then connect to a layer 3 switch Cisco WS-C3650-24PS-S to another subnet. This stackable access-layer switch can provide full coverage of wired and wireless network with its flexibility. There are four 3650 layer 3 switches in total for the whole network. Also, there are 2 Cisco 4331 Integrated Services Router for connecting over network. The throughput of 4331 router is between 100 and 300 Mbps which provides high and stable bandwidth. Moreover, there are 2 access points for wireless devices. Those components have high compatibility with other devices that allow school for further expending.



Cisco 4331
Integrated Services Router



Cisco WS-C3650-24PS-S



Cisco Catalyst 2950 Series
24-Port Fast Ethernet Switch

IP addresses allocation

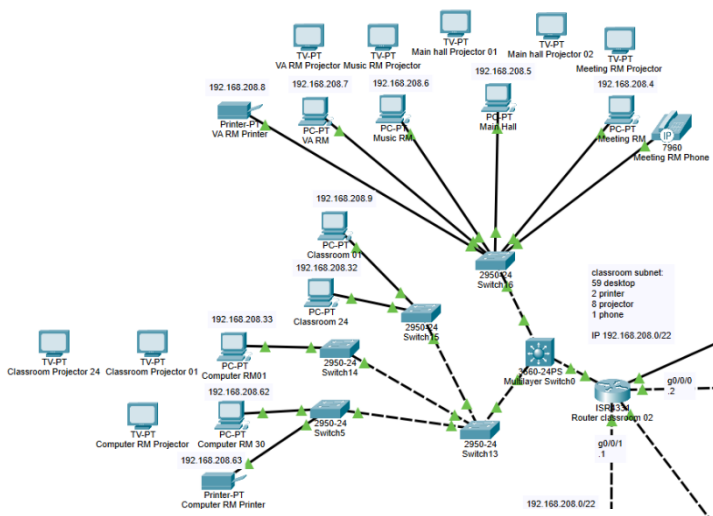
There are 162 IP addresses for existing devices (desktop, servers, printer, etc.). Providing wired and wireless service to staff and students. Static IP addresses will be used for all the wired devices. Using static IP is easy to manage the intranet. A consistent network identity can enhance security by limiting access in intranet. Also, firewall and intrusion detection system can be added to increase protection by blocking untrusted and reducing the risk of being invaded or attacked. There are 1020 dynamic IP addresses for staff and student wireless devices. Staff and students can connect to networks by different mobile devices using dynamic IP addresses. DHCP allows the network to assign IP addresses efficiently. Since not all staff or students' mobile devices will use the network at the same time, using DHCP can reuse IP addresses that can save network resources.

Subnet Areas

Subnet	Area
192.168.192.0/22	Wireless Network
192.168.208.0/22	Classrooms, Main Hall & Meeting Room
192.168.224.0/22	Library & IT server
192.168.240.0/22	Staff Offices

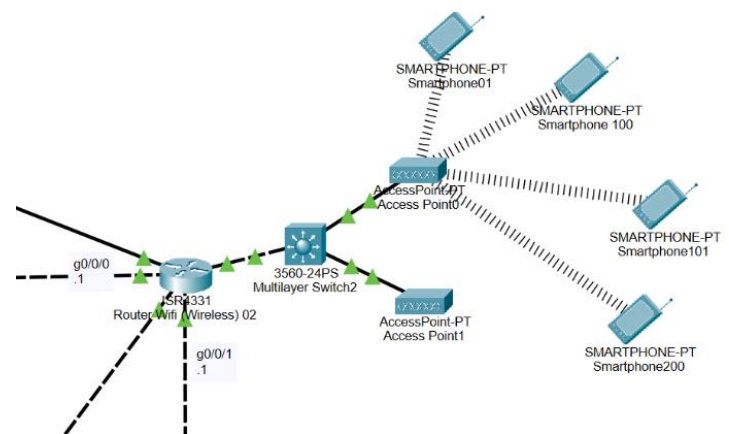
For rooms and devices, separating into 4 areas. The first area is staff, clerks and principal office. The IP address is 192.168.240.0/22. The second area is the library and IT server. The IP address is 192.168.224.0/22. The third area is the classroom and meeting room. The address is 192.168.208.0/22. The fourth area is for smartphones and portable devices. The IP address of wireless devices is 192.168.192.0/22.

The IP address ranges are from 192.168.192.1 to 192.168.195.254 (Wireless Network), 192.168.208.1 to 192.168.211.254 (Classrooms, Main Hall & Meeting Room), 192.168.224.1 to 192.168.227.254 (Library & IT server) and 192.168.240.1 to 192.168.243.254 (Staff Offices), 4088 IP address in total (excluding subnet address and broadcast address). After assigning all the nodes and routers, it will remain around 3798 IP addresses for future expanding, such as new staff or students and new devices.



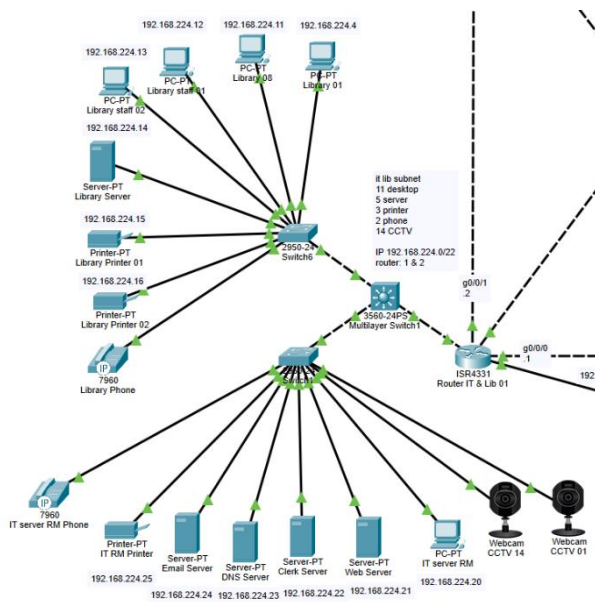
192.168.208.0/22

Classrooms, Main Hall & Meeting Room



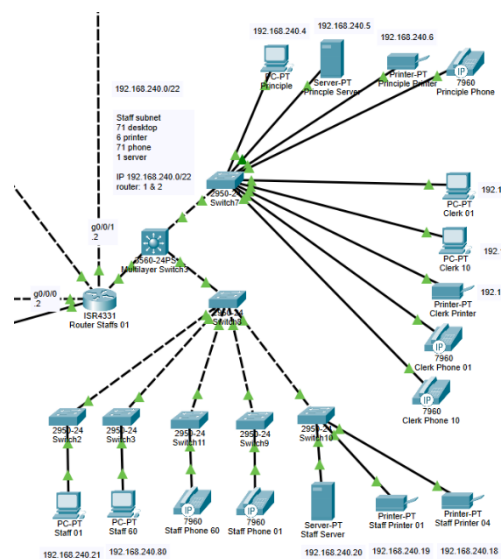
192.168.192.0/22

Wireless Network



192.168.224.0/22

Library & IT server



192.168.240.0/22

Staff Offices

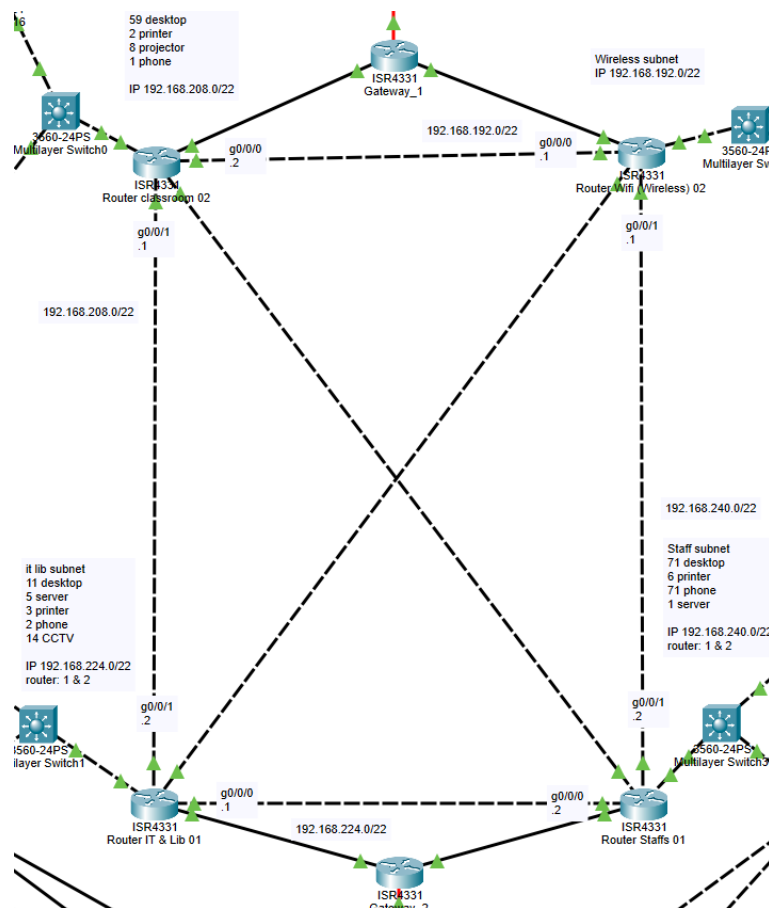
Server Application

There are 7 servers that will provide different application services. 3 servers provide web service, email service and DNS service. 4 servers provide storage for principle and staff storing teaching materials and administrative documents. Details of server service are on the table below.

Name	Record Type
speedss.edu.hk	MX
www.speedss.edu.hk, 192.168.224.23	A
speedss.edu.hk, host.speedss.edu.hk	CNAME
speedss.edu.hk, dns.speedss.edu.hk	NS

Network Topology

Mesh Topology will be used in the network set up. Routers are connected to every other in the network. They can send packets directly without forwarding them to another router to ensure the data transmit consistency. It can reduce network resources and provides an excellent deal of redundancy to ensure productivity and efficiency. Meanwhile, HSRP will set up a router which can avoid a single device failure and traffic problems. It can provide network redundancy for IP network. Ensure that user traffic is quickly and transparently switched over in case of any failure in network edge devices or access circuits. And adding new devices will not affect the network transmission which is suitable for future extension. Furthermore, port-based VLAN will be used through the 3650-24PS-S to avoid physical limitations and isolating different network groups. VLAN's network segmentation can reduce the broadcast traffic to enhance network performance and efficiency. By preventing bridge loops and broadcast storms occur, Spanning Tree Protocol is used. It disables links that are not part of spanning tree and build a loop-free topology.



For implemented the routing protocols, OSPF will be used. OSPF is available to most of the routers. It uses link-state algorithm to generate routing table which is suitable for mesh topology. It chooses the least cost or same-cost path to send packets. Using OSPF to calculate the routing cost which has different routes to access each subnet. It can prevent single point failure, link cost change and late update. Hierarchical OSPF can apply in this situation. It separates different areas that can easily for management. Besides, using DHCP to set up wireless devices. It assigns IP addresses for users that do not need to set up the IP address manually. It has strong security that all OSPF messages authenticated to prevent malicious intrusion and support multi-cast OSPF to use same topology database as OSPF. It can reduce the leakage of school's confidential documents and protect all messages.

In summary, the network infrastructure consists of Cisco Catalyst 2950 Series 24-Port Fast Ethernet Switches for each subnet, which provide wire-speed connectivity and are linked to four Cisco WS-C3650-24PS-S Layer 3 switches for comprehensive wired and wireless network coverage. Furthermore, there are two Cisco 4331 Integrated Services Routers to ensure stable bandwidths of 100-300 Mbps and two access points serving wireless devices. A mesh topology ensures direct data transmission between routers with HSRP and port-based VLANs on the 3650-24PS-S switch enhance performance through network segmentation. OSPF, using a link-state algorithm, will manage routing, ensuring efficient path selection and strong security. DHCP will dynamically assign IP addresses to wireless devices, enhancing network management and resource efficiency.

NETWORK TABLE

Devices	Number of IP addresses required	Usage of IP addresses	IP Address Allocation Method	Network IP address	Subnet Mask	Range of IP addresses	Number of IP addresses available
Wireless Network (Wi-Fi)	1010	Smartphones	DHCP	192.168.192.0	255.255.252.0	192.168.192.11 - 192.168.195.254	252
Principal's computer	1	PC	Static	192.168.240.0	255.255.252.0	192.168.240.4	1
Principal's server	1	Server	Static	192.168.240.0	255.255.252.0	192.168.240.5	1
Principal's printer	1	Printer	Static	192.168.240.0	255.255.252.0	192.168.240.6	1
Clerk's computer	10	PC	Static	192.168.240.0	255.255.252.0	192.168.240.7 - 192.168.240.16	10
Clerk's printer	1	Printer	Static	192.168.240.0	255.255.252.0	192.168.240.17	1
Staff's computer	60	PC	Static	192.168.240.0	255.255.252.0	192.168.240.21 - 192.168.240.80	60
Staff's server	1	Server	Static	192.168.240.0	255.255.252.0	192.168.240.20	1
Staff's printer	4	Printer	Static	192.168.240.0	255.255.252.0	192.168.240.18 - 192.168.240.19	2
Library's computer	8	PC	Static	192.168.224.0	255.255.252.0	192.168.224.4 - 192.168.224.11	8
Library staff's computer	2	PC	Static	192.168.224.0	255.255.252.0	192.168.224.12 - 192.168.224.13	2
Library server	1	Server	Static	192.168.224.0	255.255.252.0	192.168.224.14	1
Library printer	2	Printer	Static	192.168.224.0	255.255.252.0	192.168.224.15, 192.168.224.16	2
IT server computer	1	PC	Static	192.168.224.0	255.255.252.0	192.168.224.20	1
Web server	1	Server	Static	192.168.224.0	255.255.252.0	192.168.224.21	1
Clerk server	1	Server	Static	192.168.224.0	255.255.252.0	192.168.224.22	1
DNS server	1	Server	Static	192.168.224.0	255.255.252.0	192.168.224.23	1
Email server	1	Server	Static	192.168.224.0	255.255.252.0	192.168.224.24	1
IT room printer	1	Printer	Static	192.168.224.0	255.255.252.0	192.168.224.25	1

Devices	Number of IP addresses required	Usage of IP addresses	IP Address Allocation Method	Network IP address	Subnet Mask	Range of IP addresses	Number of IP addresses available
Meeting Room's computer	1	PC	Static	192.168.208.0	255.255.252.0	192.168.208.4	1
Main Hall's computer	1	PC	Static	192.168.208.0	255.255.252.0	192.168.208.5	1
Music Room's computer	1	PC	Static	192.168.208.0	255.255.252.0	192.168.208.6	1
VA Room's computer	1	PC	Static	192.168.208.0	255.255.252.0	192.168.208.7	1
VA Room's printer	1	Printer	Static	192.168.208.0	255.255.252.0	192.168.208.8	1
Classroom's computer	24	PC	Static	192.168.208.0	255.255.252.0	192.168.208.9 - 192.168.208.32	24
Computer Room's computer	30	PC	Static	192.168.208.0	255.255.252.0	192.168.208.33 - 192.168.208.62	30
Computer Room's printer	1	Printer	Static	192.168.208.0	255.255.252.0	192.168.208.63	1